Conceptual Framework & Model Justification

1) Students Information

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- Course: Research Methods
- Research Topic Title: CS2 Match Outcome Prediction Using Machine Learning

2) Research Question and Hypotheses

Research Question: Which machine learning model best predicts the outcome of professional CS2 matches, drawing on team performance statistics and historical match data?

Hypotheses:

- H1: Teams with higher combat performance ratings are more likely to win a given match.

- H2: A favorable head-to-head record against a specific opponent increases the probability of winning a rematch against that same team.

- H3: Teams with a strong historical performance on a given map are more likely to win when that map is selected.

3) Definition of Key Variables

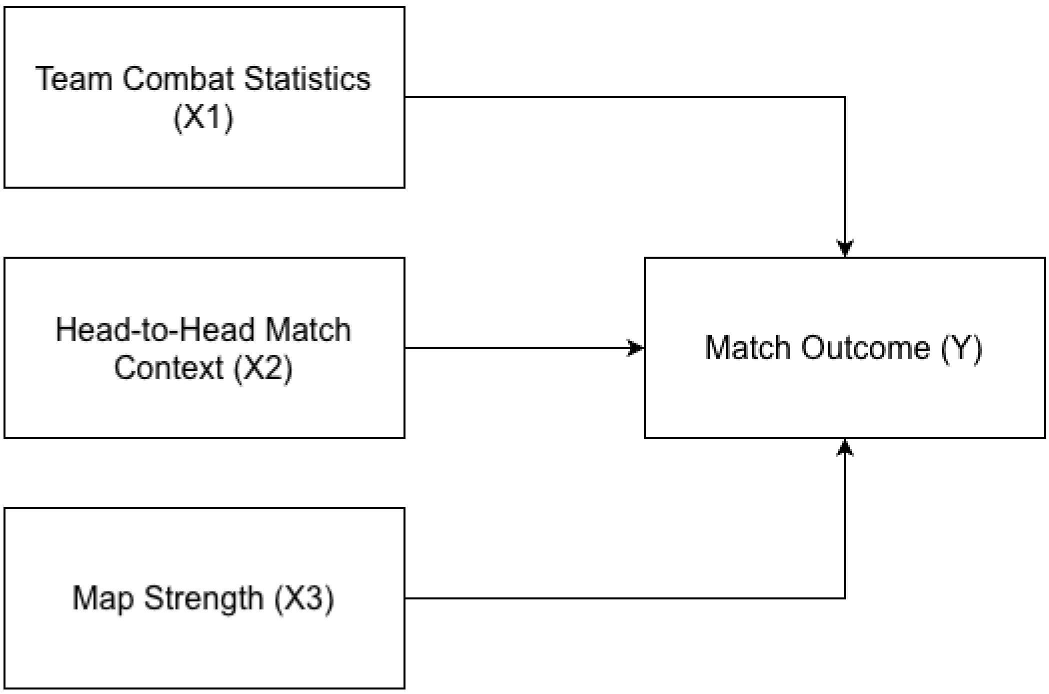
- Match Outcome
    - Role in Model: Dependent Variable (Y)
    - Conceptual Definition: The end result of a competitive CS2 match - which team wins by reaching the required number of round wins first.
    - Operational Definition: Match outcome is recorded as a binary variable - 1 for a Win, 0 for a Loss - from the perspective of the team under observation.

- Team Combat Statistics
    - Role in Model: Independent Variable (X1)
    - Conceptual Definition: How well a team fights - their ability to eliminate opponents and win engagements across a match.
    - Operational Definition: Combat performance is measured as the mean HLTV rating 2.0 and mean Kills Per Round (KPR) across the five

players on the active roster, calculated from matches played in the three months prior to the match in question.

- Head-to-Head Match Context
  - Role in Model: Independent Variable (X2)
  - Conceptual Definition: What a team's track record against a specific opponent tells us about their chances going into the next encounter - both in terms of strategy and confidence.
  - Operational Definition: The team's win rate in direct matchups against the opposing team over the past 12 months.
- Map Strength
  - Role in Model: Independent Variable (X3)
  - Conceptual Definition: How comfortable and successful a team has historically been on a given map in professional play.
  - Operational Definition: The team's win rate on the specific map being played, drawn from their professional matches in the prior 3 months using HLTV data.

4) Conceptual Framework Diagram



5) Justification of Hypothesized Relationships

- Justification for H1 (Team Combat Statistics -> Match Outcome): We expect this relationship to be positive - teams that shoot better and rack up more kills should win more often. CS2 is fundamentally about eliminating the other team; every kill makes it easier to take map control and complete objectives. Academic research in esports predictive modeling strongly supports this approach. For example, research from Lund University (2023) and Švec (2022) at the Czech Technical University demonstrate that aggregated player statistics, combat metrics, and performance ratings are foundational features for training accurate machine learning classifiers in Counter-Strike. One assumption here is that a player's recent stats fairly represent how they'll perform on match day - which isn't guaranteed, but is a reasonable starting point.

- Justification for H2 (Head-to-Head Match Context -> Match Outcome): Past results against the same opponent carry real weight. When a team has beaten a specific rival repeatedly, they're not just bringing better stats into the rematch - they're bringing a tactical blueprint that's already been proven to work, plus the psychological edge that comes with knowing they've done it before. Švec (2022) makes exactly this point: historical match records and Elo-style ratings consistently rank among the strongest predictors of match outcomes in CS:GO research. A solid head-to-head record is a signal that goes deeper than raw numbers - it reflects genuine familiarity with how a specific opponent thinks and plays.

- Justification for H3 (Map Strength -> Match Outcome): Maps in CS2 aren't interchangeable - each one has its own angles, timings, and meta, which means teams develop genuine specialisations over time. A team that consistently performs well on a particular map almost certainly has more polished utility lineups and deeper tactical preparation for it. A study from the Gabelli School of Business (2020), drawing on over 100,000 competitive matches, confirmed that map selection meaningfully shifts prediction accuracy. That accumulated experience is a real pre-match advantage - and we expect it to show up clearly in the results.

6) Model Specification

To address the central research question, we use a comparative approach rather than committing to a single model upfront. The idea is to train and test several algorithms on the same data and see which one actually performs best. The outcome variable (Match Outcome, Y) is binary - 1 for a win, 0 for a loss - and the three predictors are Team Combat Statistics (X1), Head-to-Head Match Context (X2), and Map Strength (X3).

Evaluated Models

Three machine learning models are evaluated in this study:

- Logistic Regression (Baseline Model): Logistic regression serves as the baseline. It is the most transparent of the three approaches and allows us to directly examine the contribution of each variable to the predicted win probability in a linear framework. The formula is as follows:

$$P(Y = 1) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3)}}$$

- $P(Y = 1)$ = The probability of the team winning the match
- β_0 = The intercept
- β_1, β_2, β_3 = The coefficients representing the influence of each independent variable
- X_1 = Team Combat Statistics
- X_2 = Head-to-Head Match Context
- X_3 = Map Strength

- Random Forest: This method constructs a large number of decision trees independently and aggregates their predictions. We include it because match outcomes in CS2 likely involve non-linear patterns and variable interactions that a simple logistic model won't capture - and Random Forest handles that without requiring strict assumptions about the data.

- Gradient Boosting (e.g., XGBoost): This method trains trees sequentially, where each subsequent tree is fit to the residual errors of the previous one. It is included given its strong track record on structured tabular data and its established use across sports analytics research.

Data Split Methodology

To make sure our models don't just memorize the training data and actually generalize to matches they haven't seen, we split the dataset using a standard hold-out approach:

- Training Set (80%): Used to fit all three models.
- Testing Set (20%): Held back entirely for final evaluation - none of the models see this data during training.

Model Evaluation Metrics

All three models are assessed on the test set using a consistent set of metrics to ensure comparability:

- Accuracy: The proportion of all match outcomes - both wins and losses - that the model predicts correctly.
- Precision: Among all matches the model labels as wins, the share that are genuine wins. Reflects how reliable the model's positive predictions are.

- Recall (Sensitivity): Of all actual wins in the test set, the fraction the model successfully identifies.
- F1-score: A composite measure combining Precision and Recall into a single value. Particularly relevant here due to a potential class imbalance between wins and losses in the dataset.

7) Scope and Boundaries of the Model

- Population and Context: The model is developed specifically for Tier-1 and Tier-2 professional CS2 competition, where high-quality, granular statistics are consistently accessible via platforms such as HLTV.org. Matches below this tier are excluded on the grounds that available data becomes substantially less reliable.

- What the Model Does Not Claim: Outputs from this model represent probabilistic estimates rather than definitive match forecasts. The model cannot account for situational factors that emerge immediately before a match, including roster changes, last-minute stand-in appearances, or technical disruptions.

- Limits of Generalizability: The patterns identified here are specific to the professional competitive context and do not extend to ranked matchmaking or non-professional play. Professional CS2 is a different game in terms of structure, economy management, and tactical discipline - the patterns we find here simply don't translate. On top of that, some of the most important factors in a real match - communication, mental state under pressure, in-game calls mid-round - are impossible to quantify and are left out of the model entirely.

References:

1. Erik Broms & William Nordansj¨o. *Predicting Counter-Strike Matches* (2022) Lund University Student Publications.
2. Švec, O. *Predicting Counter-Strike Game Outcomes with Machine Learning*. Faculty of Electrical Engineering (2022), Czech Technical University in Prague.
3. Henry Forgey. *Predicting Game Outcome of Counter Strike with Machine Learning: A Study of Global Offensive Competitive Matches* (2021). Fordham University.